

Project Design Phase-I

Solution Architecture

Date	03 February 2026
Team ID	LTVIP2026TMIDS24253
Project Name	Electricity-Consumption-Analysis
Maximum Marks	4 Marks

SOLUTION ARCHITECTURE

Solution architecture defines how proposed technology solution addresses the business problem of understanding and analyzing electricity consumption data across Indian states and regions. It bridges the gap between energy planning challenges and data-driven technology solutions.

Goals of Solution Architecture

- To identify most suitable technology solution for analyzing electricity consumption patterns and regional variations across 35 Indian states and 5 regions.
- To describe overall structure, components, and behavior of the system to project stakeholders including energy planners, policy makers, and researchers.
- To define key features, development phases, and functional and non-functional requirements of the solution.
- To provide clear technical specifications that guide the design, implementation, management, and delivery of electricity consumption analytics solution.

Solution Architecture Overview

The architecture employs a three-tier approach with Flask backend serving responsive frontend templates, integrating Tableau Public visualizations through iframe embedding, supported by a data layer processing 16,599 electricity consumption records with scenario-based analysis capabilities for time patterns, seasonal variations, and sector-specific consumption insights.

Technical Components

- **Frontend Layer:** HTML5, CSS3 with Bootstrap 5, and JavaScript providing responsive design and interactive user experience across all devices.
- **Backend Layer:** Python Flask framework with Jinja2 templating managing 15+ routes, business logic, and Tableau Public API integration.
- **Data Layer:** CSV file processing with validation and cleaning supporting state, regional, and temporal analysis with structured schema design.
- **Integration Layer:** Tableau Public visualization hosting with iframe embedding, URL parameter handling, and dynamic filtering capabilities.

Architecture Behavior

The system processes electricity consumption data through validation and transformation stages, delivers interactive visualizations through web interface, supports scenario-based analysis with color-coded navigation, and provides professional user experience with breadcrumb trails and responsive design patterns.

Development Phases

Phase 1 establishes foundation with core Flask application and basic Tableau integration, Phase 2 implements advanced features and analysis scenarios, Phase 3 focuses on optimization and testing, while Phase 4 delivers comprehensive documentation and deployment with professional presentation materials.

Key Features

Interactive dashboard with 12 professional visualizations, three analysis scenarios for focused insights, individual visualization pages with detailed information, responsive design for all devices, and comprehensive documentation with user guides and technical specifications.

Quality Standards

Performance optimization achieving page loads under 3 seconds, cross-browser compatibility at 95%+, security implementation with input sanitization and HTTPS encryption, and accessibility compliance with WCAG 2.1 AA standards for professional user experience.

Stakeholder Benefits

Energy planners receive data-driven insights for resource allocation, policy makers obtain evidence-based analysis for decision making, researchers access comprehensive data for academic work, and general public gains understanding of electricity consumption patterns through interactive and user-

